WasteWise: A Smart Segregation Bin

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BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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DECLARATION

We declare that this written submission represents our ideas in our own words and where others' ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented or fabricated or falsified any idea/- data/fact/source in our submission to the best of my knowledge. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been taken when needed.

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This is to certify that the report titled **"WasteWise: A Smart Segregation Bin"**, submitted by **Sheen Sarup, Ananya Goel, Shehjaar Manwati, Yuvraj Gill and Sartaj Singh,** to the Thapar Institute of Engineering & Technology, Patiala, for the award of the degree of **Bachelor of Engineering**, is a record of the project work done by them under our supervision. The contents of this report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

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ABSTRACT

This Waste management is a significant global issue, with improper disposal and segregation leading to adverse effects on the environment, human health, and resource conservation. Current methods of waste segregation, particularly manual sorting by waste collectors and rag pickers, are inefficient, inaccurate, and pose considerable health risks. The poor quality of segregated waste leads to contaminated recyclables, which drives up processing costs and undermines the effectiveness of recycling efforts. This thesis presents the development of a smart waste segregation system designed to address these challenges by automating the classification of waste at the source.

The system utilizes image processing techniques and machine learning algorithms to classify waste into three categories: organic, recyclable, and non-recyclable. Image recognition technology identifies waste based on visual characteristics, and machine learning algorithms trained on a comprehensive dataset ensure the system can accurately segregate waste, even in mixed or complex conditions. By automating this process, the system reduces human intervention, thus minimizing health risks and improving the efficiency of waste management operations. Moreover, the system enhances recycling quality by reducing contamination rates, which increases the economic viability of recycling programs.

This thesis discusses the design, implementation, and performance evaluation of the smart waste segregation system. The system is designed to be scalable and adaptable, suitable for integration into municipal waste management programs, industrial facilities, and household use. Machine learning models are continuously updated with new data, ensuring the system remains effective as waste management needs evolve.

Layout of the Thesis

Chapter 1: Introduction introduces the problem of waste management, particularly the limitations of manual segregation, and outlines the motivation for automating waste classification.

- **Chapter 2: Literature Survey** reviews existing methods and technologies for waste segregation and explores the use of automation, image processing, and machine learning in waste management.

- **Chapter 3: Problem Formulation and Objectives** details the challenges in waste segregation and sets the specific objectives for developing an automated system to classify waste efficiently.

- **Chapter 4: Project Design and Description** presents the system architecture, covering the image processing techniques and machine learning models used for waste classification. The chapter also details the hardware and software components implemented.

- **Chapter 5: Outcomes and Prospective Learning** discusses the system's performance, including its accuracy, scalability, and potential for integration into various waste management scenarios. Lessons learned during the project development are also outlined.

- **Chapter 6: Project Timeline** provides a timeline of the project's development, listing key milestones and individual plans for each phase.

- **Chapter 7: Conclusion and Future Work** summarizes the system's contributions to waste management, discusses its potential for future improvements, and suggests directions for further research.

Keywords: Waste segregation, automation, machine learning, image processing, recycling, smart system, waste management.

This abstract highlight the key features and outcomes of the smart waste segregation system, offering a comprehensive overview of the project's significance in enhancing waste management practices.

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LIST OF ABBREVIATIONS

TIET	Thapar Institute of Engineering and Technology
ECE	Electronics and Communication Engineering
IoT	Internet of Things
AI	Artificial Intelligence
DSP	Digital Signal Processing
YOLO	You Only Look Once
OpenCV	Open-Source Computer Vision Library
Wi-Fi	Wireless Fidelity
mAP	Mean Average Precision
IEEE	Institute of Electrical and Electronics Engineers
LoRa	Long Range
GUI	Graphic User Interface
ML	Machine Learning

CHAPTER 1
INTRODUCTION

The project "WasteWise: A Smart Segregation Bin" aims to develop an automated waste management system using machine learning and computer vision. The project addresses the need for efficient waste management, resource recovery, and pollution reduction by employing machine learning and computer vision. The key highlights of the project are as follows :-

- Uses Raspberry Pi and robotic automation to classify waste.
- Enhances public awareness and automates waste classification into categories like organic, recyclable, and non-recyclable.
- Integrates machine learning techniques and image processing for waste segregation.
- Identifies gaps in current systems, such as the need for more efficient classification into specific categories (organic, recyclable, non-recyclable).

The project aims to create a cohesive system for automated waste segregation, leveraging modern technologies to improve waste management efficiency and sustainability.

1.1 OBJECTIVES AND SCOPE

Objectives: The primary objective of this project is to develop a smart waste segregation system that automates the classification of waste into three categories: organic, recyclable, and non-recyclable. The specific objectives are as follows:

1. **Automate waste segregation:** Design and implement a system that uses image processing and machine learning algorithms to accurately identify and classify different types of waste.
2. **Improve segregation efficiency and accuracy:** Ensure that the system achieves high accuracy in identifying waste types, even in mixed or complex waste environments, to reduce errors in sorting.
3. **Reduce human intervention:** Minimize the reliance on manual waste sorting by automating the process, thus lowering the health risks faced by waste collectors and rag pickers.
4. **Enhance recycling quality**: Improve the quality of recyclable materials by reducing contamination, leading to more effective recycling processes and reduced processing costs.
5. **Scalability and adaptability:** Design the system to be scalable for use in various environments, including municipal waste programs, industrial waste management, and household waste segregation.
6. **Continuous learning:** Incorporate machine learning models that can adapt and improve with new data, ensuring the system remains effective as waste management needs evolve.

Scope:

1. **Target Users:** Designed for individuals and organizations involved in waste management, including households, municipal corporations, industrial facilities, and waste collection agencies. The system is particularly beneficial for areas struggling with inefficient waste segregation and recycling.

2. **Customization:** The smart waste segregation system features modular design components, allowing it to be customized based on the volume and type of waste generated. It can be adapted for use in small households, large industrial complexes, or municipal waste collection centers.

3. **Cost-Effectiveness:** The system aims to provide an affordable and sustainable waste management solution, reducing labor costs and improving recycling efficiency, thereby making it accessible to a wide range of users, including municipalities with limited budgets.

4. **Versatility of Use:** Suitable for deployment in diverse environments, including residential areas, offices, industrial plants, and public spaces. It can also be integrated into existing waste management infrastructures or used as a standalone solution.

5. **Technology Integration:** The system incorporates advanced image processing, machine learning algorithms, and real-time classification techniques. It is designed to adapt to new waste types through continuous learning, ensuring long-term effectiveness and improving waste segregation accuracy over time.

6. **Safety and Compliance:** Developed to meet environmental and operational safety standards, the system ensures that it operates reliably in various environments without posing risks to users or the environment. It is also designed to reduce health risks for waste handlers by automating the waste segregation process.

CHAPTER 2
LITERATURE SURVEY

1. A Method for Waste Segregation using Convolutional Neural Networks", by Jash Shah and Sagar Kamat

The paper, concludes that global increase in waste generation has overwhelmed the garbage and recycling industries, necessitating smart solutions for improved environmental monitoring and recycling processes. Waste disposal impacts human life and the environment, but effective waste management systems can mitigate these negative effects. Waste classification can be either manual, relying on human effort, or automated using various techniques. As recycling becomes vital for sustainability, the hidden costs of sorting and processing recyclable materials pose challenges. Despite consumer efforts to sort waste, confusion often arises in categorizing diverse items, highlighting the need for automated recycling solutions that offer environmental and economic benefits.

Landfilling organic waste is problematic due to methane production during anaerobic decomposition, which significantly contributes to greenhouse gas emissions. Improperly managed organic waste can contaminate water sources and promote harmful bacteria and fungi growth. Most municipal waste, not segregated at the source, ends up in landfills. Advanced systems using computer vision and AI, such as convolutional neural networks (CNN), can improve the sorting of recyclable materials, reducing landfill waste. This study presents three models—ResNet-34, VGG16, and a proposed deep neural network—for image classification, segmentation, and detection, aiming for a more accurate and optimal waste categorization method for industrial applications.[1]

2. Object Detection-based Automatic Waste Segregation using Robotic Arm

The paper, "Object Detection-based Automatic Waste Segregation using Robotic Arm", by Azza Ibrahim, et. al., presents a smart waste segregation system using a robotic arm and the YOLOv6 object detection model to efficiently sort garbage in real time. The system addresses challenges of waste management exacerbated by overpopulation and urbanization. YOLOv6's hardware-friendly design ensures high detection accuracy and low inference time, while a 4-DOF robotic arm, designed using CAD software, utilizes a geometric method for precise joint angle calculations, enhancing

performance. Tested on a modified TrashNet dataset with seven garbage classes, the system outperforms recent models in precision, recall, F1 score, and processing speed, offering a practical solution for automatic waste segregation.[2]

3. ConvoWaste: An Automatic Waste Segregation Machine Using Deep Learning

The research paper presents an innovative solution for urban waste management through automated waste segregation. The proposed system employs a deep convolutional neural network (DCNN) and image processing techniques to classify six types of waste: plastic, metal, glass, organic, medical, and electronic. The machine utilizes a servo motor-based system to place detected waste into corresponding bins and includes an ultrasonic sensor and GSM-based communication for real-time monitoring and notifications. Controlled remotely via an Android app, the system aims to enhance recycling efficiency and resource optimization, contributing to a circular economy. The ConvoWaste model achieved a 98% accuracy rate, demonstrating significant potential for practical applications in sustainable waste management.[3]

2.2 RESEARCH GAPS

1. **EFFICIENT CATEGORISATION**: A majority of the research papers analysed used either rudimentary categories of dry/wet waste; or complicated the sorting process by sorting each material at source. The team working on WasteWise found this to be an inefficient approach. Instead, it was deemed appropriate to sort the waste into three categories: Recyclable, Non-Recyclable, and Organic Waste. This allows for an efficient sorting, while also ensuring that the classification does not cumbersome. The development process of the model found it to be highly accurate and efficient.

2. **MIXING OF WASTE:** It was also observed that the models used in the research paper thus analysed did not address the problem of two or more types of waste mixed and discarded. This is a problem area that persists with the WasteWise model as well.

3. **RESOURCE ALLOCATION:** Furthermore, it was found that a variety of papers used varieties of Raspberry Pi Model 3 and Arduinos in combination to detect and classify the objects and operate the motors for the bins accordingly. This leads to an increase in costs. It also wastes resources in integration of the Arduinos and Raspberry Pi. The model of WasteWise addresses this issue by proposing to use a Raspberry Pi 4 Model B. The 8 GB memory ensures sufficient storage for object detection, classification and operating the 4 Servo motors as well.

2.3 PROBLEM DEFINITION AND SCOPE

1. **PROBLEM DEFINITION:** The increasing volume of waste and improper segregation pose significant environmental challenges, leading to inefficient recycling processes and pollution. Manual sorting of waste is labor-intensive, prone to errors, and exposes workers to hazardous conditions. An automatic waste segregation system is needed to streamline waste management, ensure accurate classification of waste into recyclable, non-recyclable, and organic categories, and reduce human intervention. This can significantly enhance the efficiency of recycling programs, minimize environmental impact, and promote sustainable waste management practices.

2. **SCOPE:** This project aims to develop an automatic waste segregation bin utilizing a Raspberry Pi and a Pi camera module for real-time image capture and processing. The system will employ image classification algorithms in Roboflow to categorize waste into recyclable, non-recyclable, or organic sections. Servo motors will be integrated to direct the waste to the appropriate bin sections based on the classification results. The scope includes designing the hardware setup, developing the image processing and classification software, integrating the components, and testing the system to ensure accuracy and reliability in various waste segregation scenarios. This project targets applications such as households and public spaces, such as in college campuses and office spaces.

CHAPTER 3
PROBLEM FORMULATION AND OBJECTIVES

3.1 PROBLEM FORMULATION

The escalating challenge of waste management in modern societies is a multifaceted problem driven by rapid urbanization, population growth, and increased consumerism. As the volume of waste generated globally continues to rise, traditional waste management practices struggle to keep pace. One of the critical issues is the ineffective segregation of waste at the source, which is essential for efficient recycling, resource recovery, and minimizing environmental pollution.

Manual sorting methods currently dominate waste segregation processes. These methods are not only labor-intensive and costly but also prone to human error and inconsistencies. Workers are often exposed to hazardous materials, leading to health risks. Moreover, the lack of proper segregation results in recyclable and organic waste ending up in landfills or incinerators, contributing to soil degradation, water contamination, and greenhouse gas emissions. Public awareness about proper waste disposal is generally low, and existing waste bins do not facilitate or encourage correct segregation. The absence of user-friendly and automated systems means that individuals are less likely to sort their waste appropriately, exacerbating the problem.

Technological advancements in machine learning, computer vision, and robotics present an opportunity to revolutionize waste management systems. However, current implementations are either too expensive, lack scalability, or do not integrate seamlessly into everyday public use. There is a significant gap between technological potential and practical, accessible solutions that can be widely adopted.

WasteWise: A Smart Segregation Bin project seeks to bridge this gap by developing an automated waste segregation system that is both efficient and user-friendly. By harnessing the power of Raspberry Pi, machine learning algorithms, and robotic automation, the project aims to accurately classify waste into specific categories—organic, recyclable, and non-recyclable—in real-time. The system employs image processing techniques to identify waste items, reducing the reliance on manual sorting and minimizing errors.

3.2 OBJECTIVES

The main objectives of this project are as follows:

1. **Develop an Automated Waste Segregation System:** Create a smart waste bin that utilizes machine learning algorithms and computer vision to automatically classify waste.
2. **Implement Advanced Technologies:** Integrate Raspberry Pi and robotic automation to physically sort waste into organic, recyclable, and non-recyclable categories.
3. **Enhance Waste Classification Accuracy:** Employ image processing techniques to improve the precision of waste identification and segregation in real-time.
4. **Address Gaps in Current Waste Management Systems:** Provide a more efficient and specific classification system compared to existing methods, targeting the limitations of manual sorting.
5. **Promote Public Awareness and Engagement:** Increase public understanding of proper waste disposal practices through the deployment of smart segregation bins in communities.
6. **Improve Environmental Sustainability:** Contribute to pollution reduction and resource recovery by ensuring waste is correctly sorted and processed for recycling or proper disposal.
7. **Create a Scalable and Cohesive Solution:** Design the system to be scalable and adaptable for widespread adoption in various settings, such as residential areas, commercial spaces, and public venues.
8. **Leverage Machine Learning for Continuous Improvement:** Implement algorithms that learn and improve over time, enhancing the system's ability to accurately classify emerging types of waste.
9. **Facilitate Data Collection and Analysis:** Collect data on waste types and volumes to inform waste management strategies and policies.
10. **Encourage Sustainable Practices:** Support the transition towards more sustainable waste management by integrating technology that simplifies and automates the segregation process.

CHAPTER 4
PROJECT DESIGN AND DESCRIPTION

4.1 DESCRIPTION

The project "WasteWise: A Smart Segregation Bin" is dedicated to developing an innovative automated waste management system utilizing machine learning and computer vision technologies. With a focus on enhancing waste management efficiency, resource recovery, and pollution reduction, WasteWise aims to address the pressing need for sustainable waste solutions.

Key Features:

1. Utilizes Raspberry Pi and robotic automation for waste classification.

2. Raises public awareness and streamlines waste classification into categories such as organic, recyclable, and non-recyclable.

3. Integrates machine learning techniques and image processing algorithms for efficient waste segregation.

4. Identifies and addresses gaps in current waste management systems, particularly in the classification of specific waste categories.

WasteWise endeavours to create a comprehensive system for automated waste segregation, leveraging modern technologies to revolutionize waste management practices. By harnessing the power of machine learning and computer vision, this project strives to optimize waste sorting processes, minimize environmental impact, and promote sustainability.

4.2 SYSTEM ARCHITECTURE

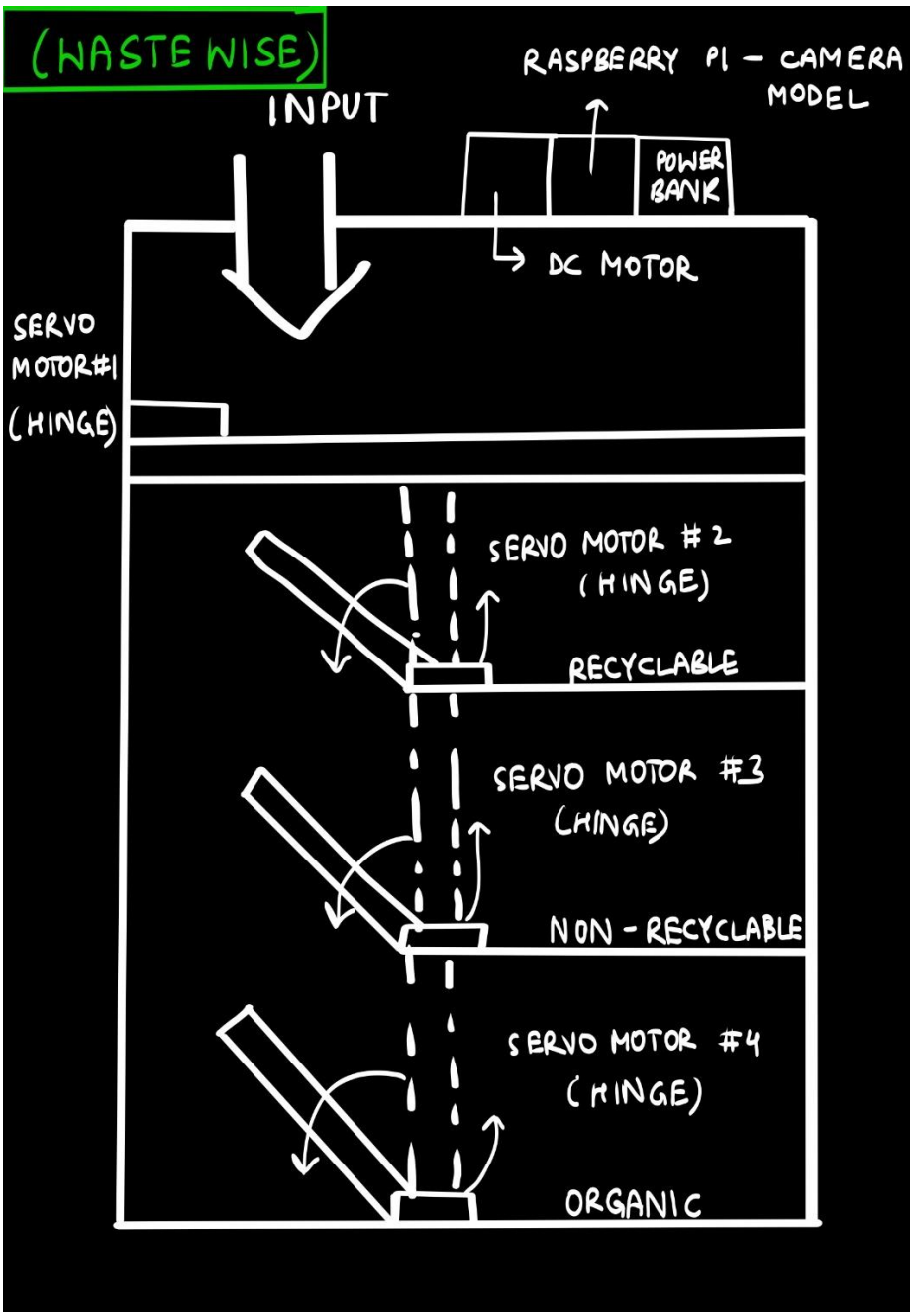


Figure:1 (Project Architecture)

4.2.1 HARDWARE DESIGN

Figure:2 (Final Working Prototype)

1. **Raspberry Pi with Camera Module:** Acts as the brain of the system, capturing images of waste and processing them through machine learning algorithms for classification into recyclable, non-recyclable, and organic categories.
2. **DC Sliding Motor:** Located at the top of the bin, this motor moves forward to hold the waste once it's placed into the input section. After classification, the motor releases the waste, allowing it to fall into the appropriate bin.
3. **Servo Motors (Hinge Mechanism):**
    - **Servo Motor 2,3:** Directs recyclable waste into its compartment.
    - **Servo Motor 1,4:** Moves non-recyclable waste to its assigned bin.
4. **Power Bank:** Supplies power to the Raspberry Pi, DC sliding motor, and servo motors, ensuring smooth operations.
5. **Outer Wooden Structure:** The outer casing of the bin is made from wood, which gives the system added stability and durability.

4.2.2 WORKING MECHANISM

The **Smart Waste Segregation Bin** automates the process of classifying and sorting waste into three distinct categories: recyclable, non-recyclable, and organic waste. This is achieved through a combination of image processing, machine learning, and mechanical sorting mechanisms. The following steps outline how the system operates from the moment waste is placed into the bin until it is sorted into the appropriate section.

Step 1: Waste Input

The waste is placed in the input section of the bin, where a **DC sliding motor** located at the top of the bin moves forward to hold the waste securely. This motor ensures that the waste remains in place during the classification process, preventing any misplacement.

Step 2: Image Capture and Classification

Once the waste is secured by the DC motor, the **Raspberry Pi** and **camera module** take an image of the waste. The Raspberry Pi processes the image using a trained machine learning model that classifies the waste into one of three categories:

- **Recyclable waste** (e.g., plastic, paper)
- **Non-recyclable waste** (e.g., hazardous waste)
- **Organic waste** (e.g., food waste)

The machine learning model has been trained to recognize different waste types based on their visual characteristics, such as shape, colour, and texture.

Step 3: Sorting Mechanism

Once the waste is classified, the DC motor releases the waste, allowing it to fall downwards. At this point, one of the four **servo motors** (hinge-based) is activated to direct the waste into the appropriate compartment:

- **Servo Motor 2,3:** Directs recyclable waste into its compartment.
- **Servo Motor 1,4:** Moves non-recyclable waste to its assigned bin.

The precise control of the servo motors ensures that the waste is sorted efficiently and with minimal error.

Step 4: Emptying the Bins

To simplify the process of emptying the bins, the system features a **detachable side log** in each compartment. This log can easily be removed, providing convenient access to the waste inside. When the sensors indicate that a section is full, the user can simply detach the side log to empty the corresponding compartment without disturbing the other sections. This design enhances the usability of the bin, making it easy to maintain and clean.

4.2.3 SOFTWARE USED

1. **Raspberry Pi OS**
    - **Role**: Acts as the operating system for the Raspberry Pi, providing a platform for running the machine learning models, controlling hardware like servo motors, and managing the overall software stack.
    - **Why it's used**: It's lightweight and optimized for Raspberry Pi, enabling smooth integration with hardware components and software libraries.
2. **TensorFlow**
    - **Role**: Used for developing, training, and deploying machine learning models, specifically for image classification.
    - **Why it's used**: TensorFlow offers powerful tools for machine learning, allowing the team to train models for accurately classifying waste as organic, recyclable, or non-recyclable.
3. **Roboflow**
    - **Role**: A platform for dataset preparation, annotation, and augmentation.
    - **Why it's used**: Roboflow simplifies the process of curating datasets by providing tools for image labelling and augmentation, essential for improving the accuracy of the machine learning models used in waste classification.
4. **YOLOv5 (You Only Look Once Version 5)**
    - **Role**: Used for real-time object detection in the waste classification process.
    - **Why it's used**: YOLOv5 is known for its speed and accuracy in object detection, making it an ideal choice for identifying waste types in real-time.

5. **OpenCV (Open-Source Computer Vision Library)**
    - **Role**: Handles image processing tasks, such as detecting objects and extracting features from the waste images.
    - **Why it's used**: OpenCV is widely used for real-time computer vision applications, making it suitable for processing the images captured by the Raspberry Pi camera.
6. **Python**
    - **Role**: The primary programming language for developing the software components of the project.
    - **Why it's used**: Python provides extensive libraries for machine learning (TensorFlow), image processing (OpenCV), and hardware integration with Raspberry Pi, making it versatile for developing the entire system

4.3 STANDARDS USED

This project incorporates several IEEE standards to ensure compatibility, reliability, and efficiency in the system's design and operation. The following standards were applied throughout various components of the WasteWise system:

1. **IEEE 1451 – Smart Transducer Interface for Sensors and Actuators:** IEEE 1451 defines a set of standards for interfacing smart sensors and actuators with networks. In the context of this project, IEEE 1451 supports the communication between sensors (such as those used for detecting waste levels) and the Raspberry Pi. By adhering to this standard, the project ensures that data from sensors is transmitted in a standardized format, facilitating accurate integration, data acquisition, and remote monitoring. This standard also aids in the modularity of the system, allowing for future upgrades or additional sensors to be integrated seamlessly.

2. **IEEE 802.11 – Wi-Fi Communication:** IEEE 802.11 standardizes Wi-Fi communication protocols, which were essential in enabling the Raspberry Pi to connect to a network for data transmission and real-time monitoring. Through this standard, the project can support wireless connectivity, allowing users to monitor bin status remotely. This setup could be particularly useful in municipal applications where waste collection schedules can be optimized based on the real-time status of each bin.

3. **IEEE 1588 – Precision Time Protocol (PTP):** IEEE 1588 defines the Precision Time Protocol (PTP) for accurate clock synchronization across a network, which is crucial in applications that require coordinated actions among multiple components. In this project, PTP ensures synchronization between image capture and processing on the Raspberry Pi, facilitating precise timing in waste classification and bin sorting. This time synchronization helps optimize the system's response time and operational accuracy, particularly when managing waste in real-time and reducing delays in servo motor activation.

4. **IEEE 802.3 – Ethernet for Wired Communication:** While the project primarily utilizes Wi-Fi (IEEE 802.11), IEEE 802.3 (Ethernet) is available as a wired communication backup. This Ethernet standard provides a reliable, high-speed data transfer option, which could be useful in environments where stable Wi-Fi is unavailable or where high-speed data transmission is necessary. In case of network outages, the Ethernet connection ensures that the data transmission between the Raspberry Pi and any external monitoring systems remains uninterrupted.

CHAPTER 5
OUTCOME AND PROSPECTIVE LEARNING

5.1 PROJECT SUMMARY

This project focuses on the design and development of an automated waste segregation system, targeting the challenges of improper waste disposal and inefficient recycling processes. The "WasteWise" smart bin leverages machine learning and computer vision to classify waste into three categories—organic, recyclable, and non-recyclable—thereby reducing the need for manual sorting. Using a Raspberry Pi 4 as the core processor, the system captures real-time images of waste via a camera module, processes them through advanced object detection algorithms, and automatically sorts the waste using integrated servo motors.

Key technologies include the YOLOv5 object detection model for waste classification, the TensorFlow framework for training deep learning models, and the OpenCV library for image processing. The servo motors are activated based on classification results, physically directing waste into the appropriate bin sections. The system is designed to be cost-effective, scalable, and environmentally sustainable, with potential applications in households, campuses, and public spaces.

Key aspects of the project include the integration of affordable hardware for wide accessibility, real-time waste classification for enhanced recycling efficiency, and automated segregation to reduce human intervention. The project aims to revolutionize waste management practices by offering an intelligent, data-driven solution that promotes sustainability and reduces pollution.

5.2 KEY OUTCOMES

1. **Accurate Waste Classification:** The project successfully developed a machine learning model that classifies waste into three main categories: Recyclable, Non-Recyclable, and Organic. This classification was achieved using over 800 annotated images and the YOLOv5 Deep Learning Algorithm for object detection. The model evaluation metrics such as Mean Average Precision, precision and recall demonstrate the model's effectiveness in accurately classifying waste items.

2. **Improved Recycling Efficiency:** By segregating waste more precisely, WasteWise has the potential to enhance recycling processes, reducing contamination in recycling streams. This leads to higher quality recycled materials and lowers the cost of waste processing.
3. **Real-time Waste Segregation:** The system operates in real-time, capturing images of waste using a Raspberry Pi Camera Module, processing them with machine learning algorithms, and directing waste into designated compartments via servo motors. This reduces human intervention and automates the segregation process.
4. **Sustainability and Resource Recovery:** By improving the efficiency of waste segregation, the system contributes to environmental sustainability, minimizing waste sent to landfills, and recovering valuable materials for reuse.
5. **Deployment Feasibility:** The use of cost-effective hardware such as the Raspberry Pi and image sensors makes the system scalable and deployable in various settings, including public spaces and educational campuses.

5.3 IMPLEMENTATION AND EXPERIMENTAL RESULT

5.3.1 Simulation Result

Roboflow is used for the Data collection, Analysis and Training of the model utilizing YOLOv5 Deep Learning Algorithm for object detection in real time.

We have annotated more than 800 images of waste for our dataset into Recyclable, Non-Recyclable and Organic waste.

Dataset is split into 70% train,20% valid and 10% test set.

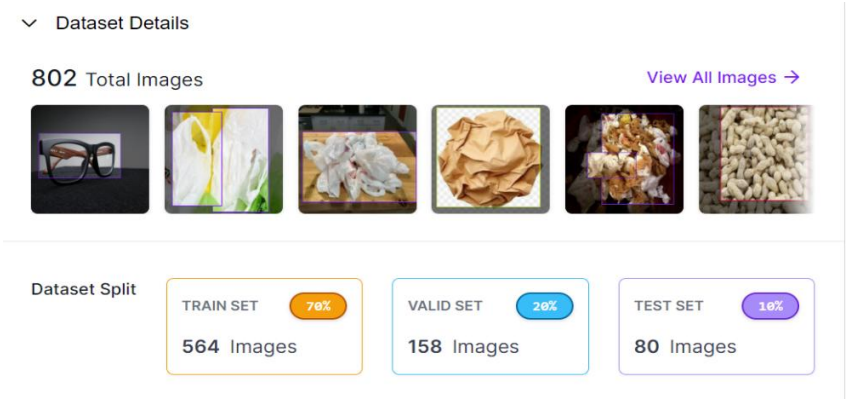


Figure:3 (Dataset Details)

- **Model Evaluation:** Here are the parameters of the mAP, Precision and Recall of the model obtained after training.
- **Precision:** It evaluates the accuracy of the positive predictions made by the model. Precision refers to the number of true positives divided by the total number of positive predictions. **Obtained Precision – 89.8%**
- **Recall:** It measures how often the model correctly identifies true positives from all the actual positive samples in the dataset. Recall is the number of true positives divided by the number of positive instances. **Obtained Recall – 82.8%**

mAP: It stand for Mean Average Precision; it is a metric which is used to evaluate the performance of the model. The mAP score ranges from 0 to 1, where a higher score indicates better model performance. It is calculated by taking the mean of the Average Precision (AP) values for each class or query. Average Precision is a way to summarize the Precision-Recall curve into a single value representing the average of precision scores at different recall levels. **Obtained mAP – 82.3%**

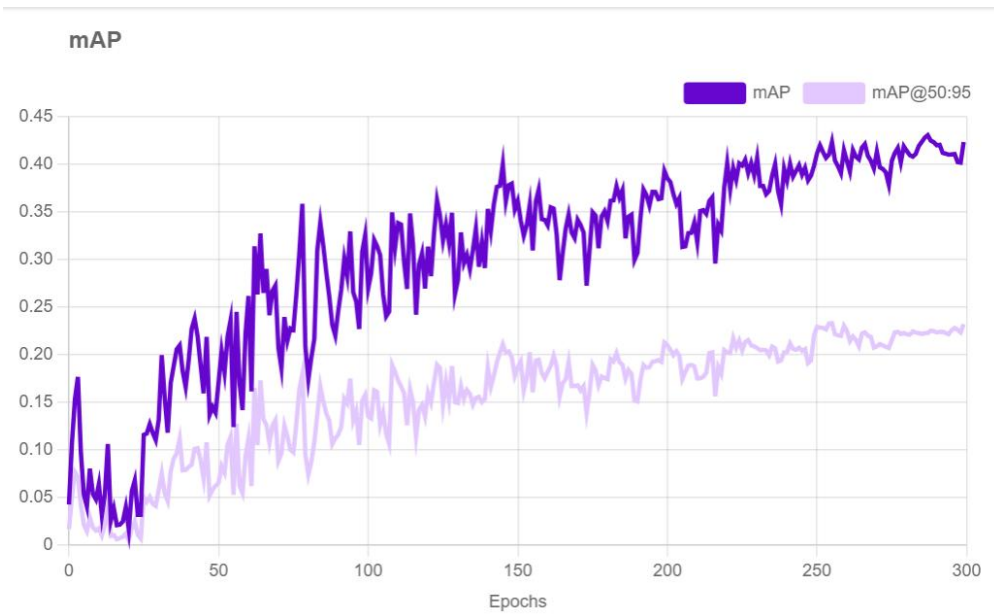


Figure:5 (Mean Average Precision)

Loss parameters of the model: The loss function, also known as the error function, quantifies the difference between the predicted outputs of a machine learning algorithm and the actual target values.

Here are the loss parameters of the model for Box Loss, Class Loss and Object Loss:

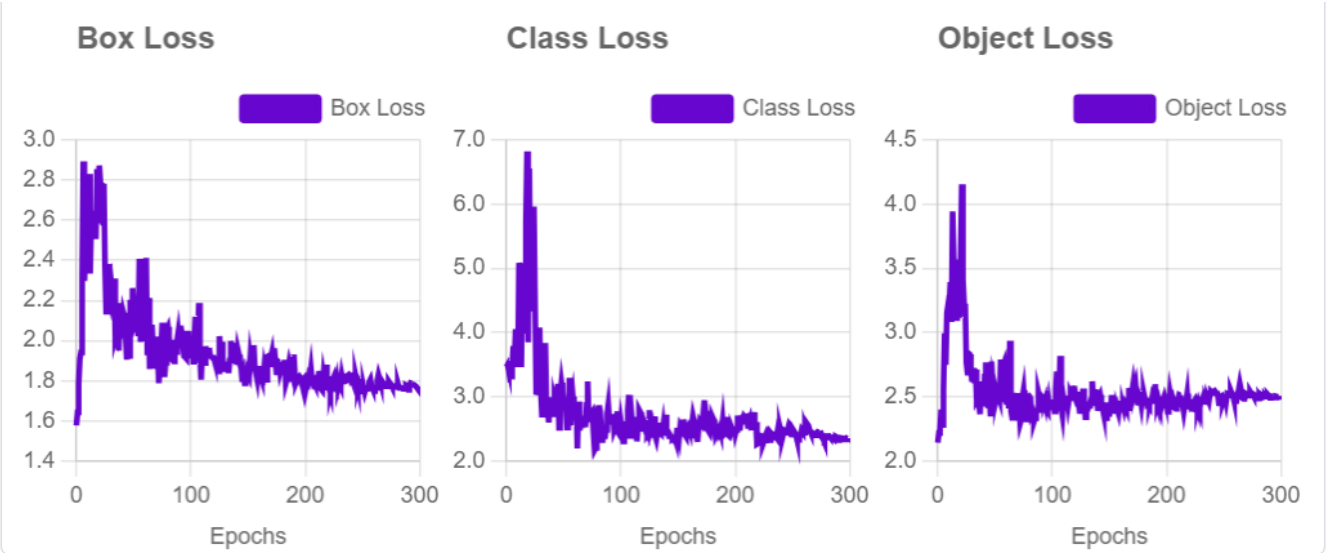


Figure:6 (Loss Parameters)

Evaluating the Loss parameters:

1. **Box Loss (Bounding Box Regression Loss):** Box Loss quantifies how well the predicted bounding boxes align with the ground truth bounding boxes. It measures the error between the predicted bounding box coordinates (e.g.-top-left corner, width, height) and the actual ground truth bounding box coordinates.

2. **Class Loss (Classification Loss):** Class Loss evaluates the correctness of class predictions made by the model for each bounding box. It quantifies the difference between the predicted class probabilities and the true class labels (whether an object belongs to a specific category or is background).

3. **Object Loss (Total loss or Multi-Task Loss):** Object loss combines both Box Loss and Class Loss to guide the overall optimization of the object detection model. It represents the overall discrepancy between predicted bounding boxes and class probabilities and the ground truth.

5.3.2 Outcomes

Classification of Recyclable objects:

As per our code and training model, these items have been correctly classified as recyclable.

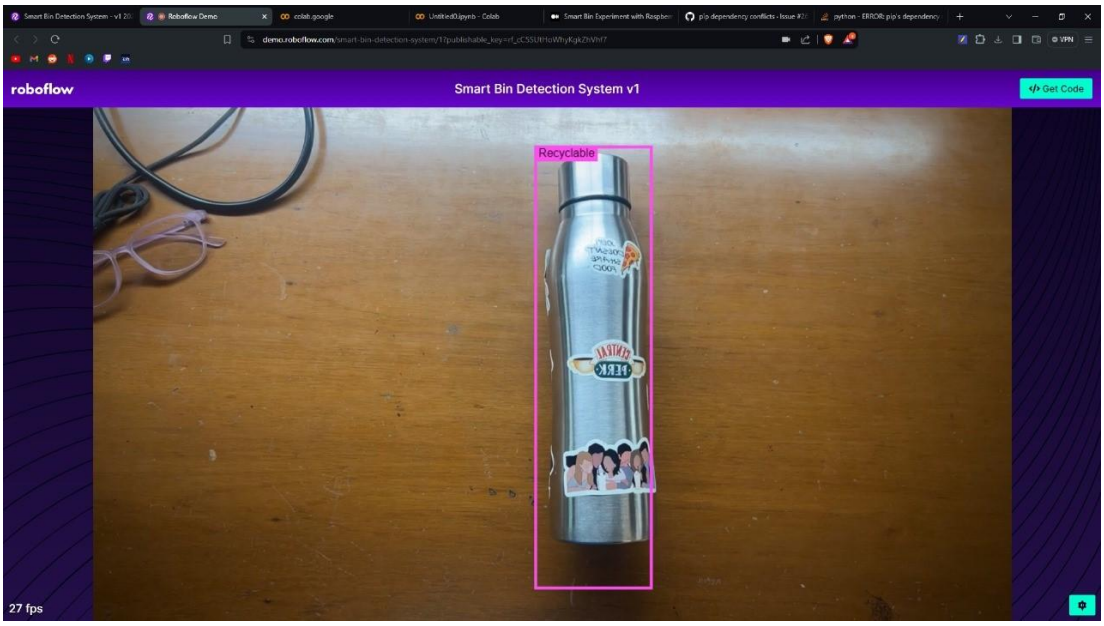


Figure:7 (Object Recognition: Recyclable)

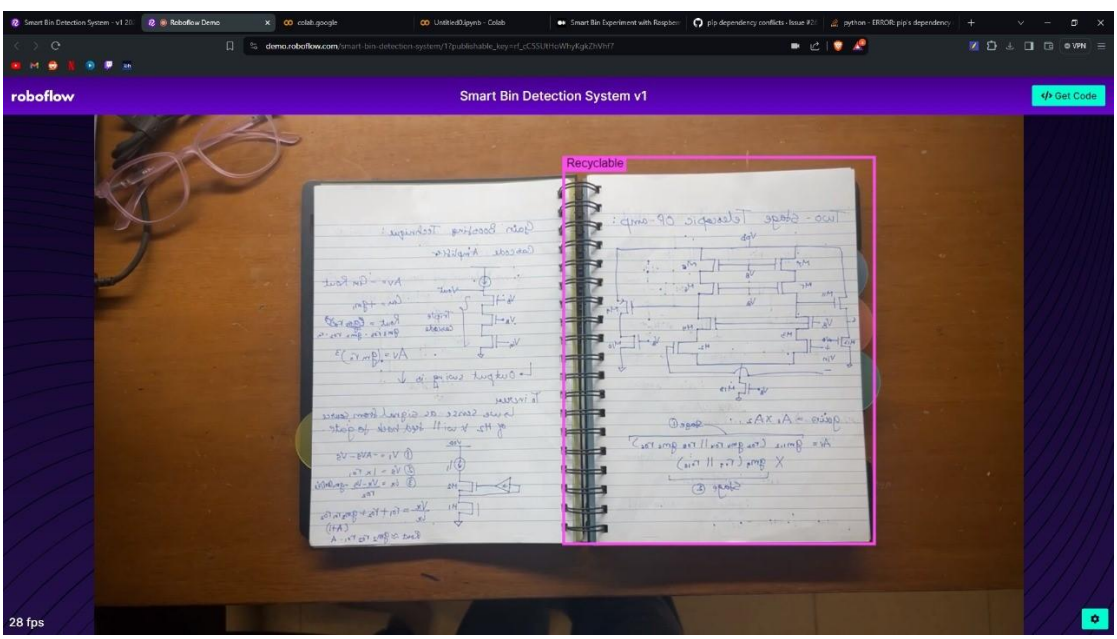


Figure:8(Object Recognition: Recyclable)

Classification of non-recyclable objects:

As per our code and training model, these items have been correctly classified as non-recyclable.

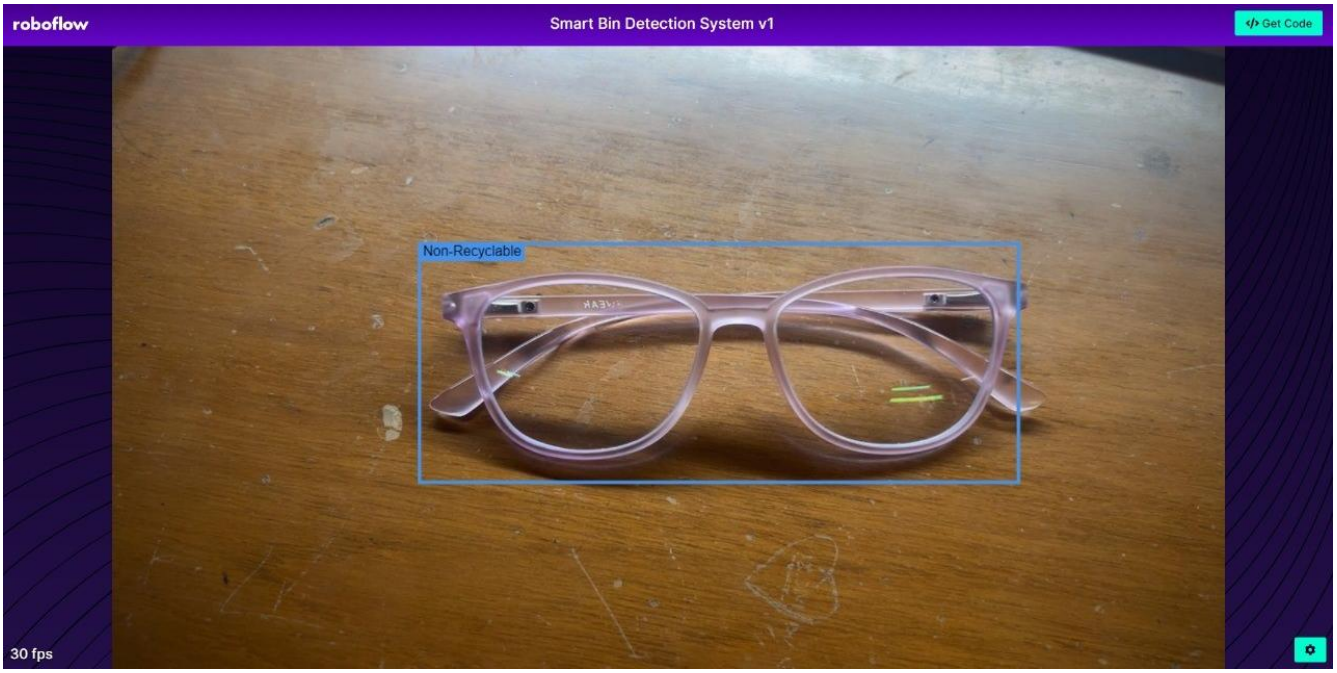


Figure:9 (Object Recognition: Non-Recyclable)

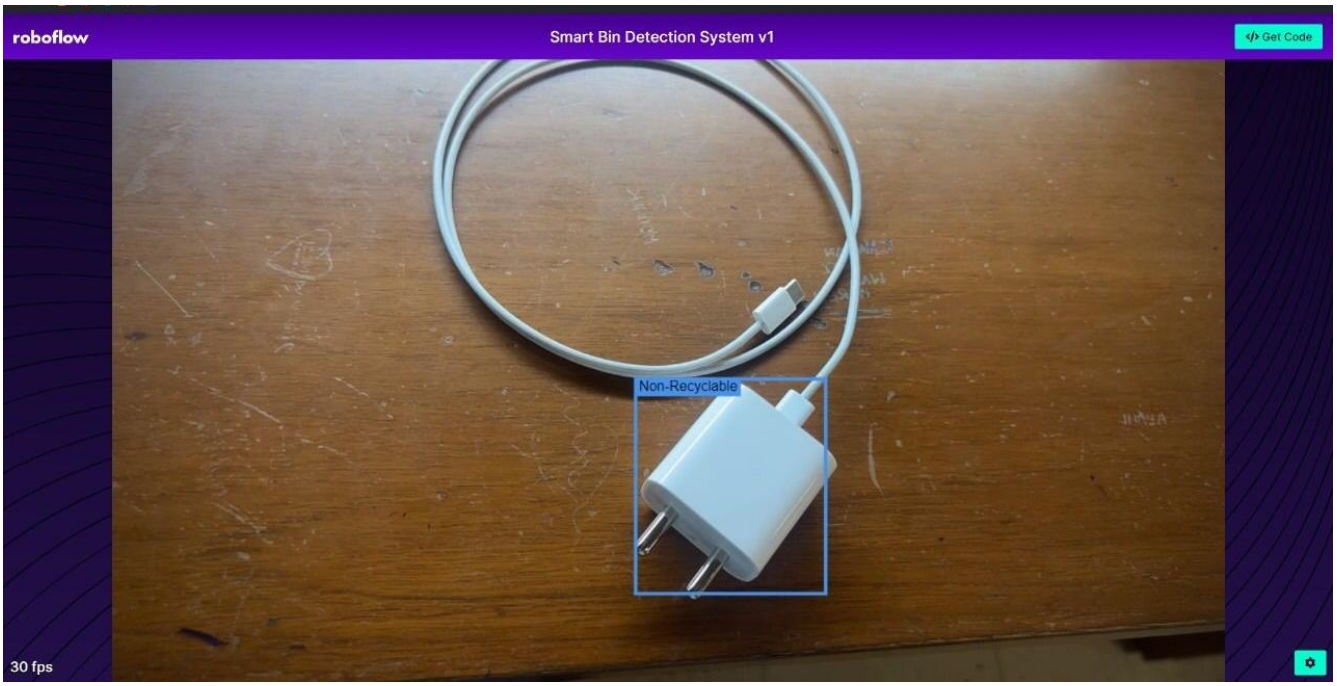


Figure:10 (Object Recognition: Recyclable)

Classification of Organic waste:

As per our code and training model, these items have been correctly classified as organic.

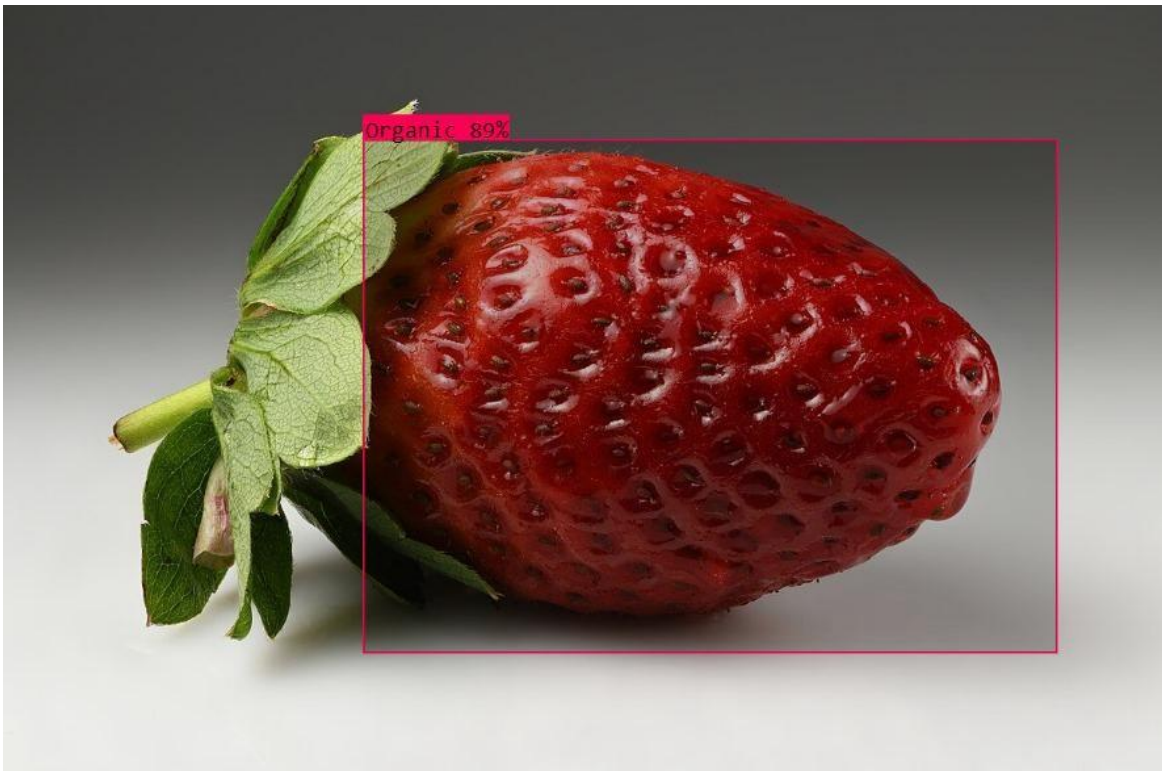
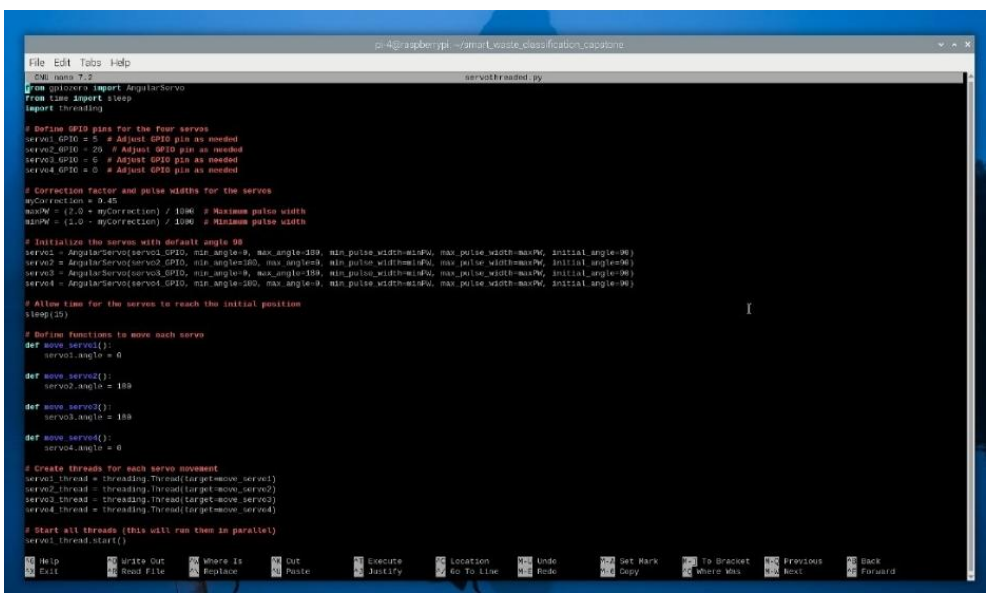


Figure:11 (Object Recognition: Organic)

Figure:12 (Object Recognition: Organic)

Figure:13 (Computing setup with Raspberry Pi)



```python
# Create threads for each servo movement
servo1_thread = threading.Thread(target=move_servo1)
servo2_thread = threading.Thread(target=move_servo2)
servo3_thread = threading.Thread(target=move_servo3)
servo4_thread = threading.Thread(target=move_servo4)

# Start all threads (this will run them in parallel)
servo1_thread.start()
servo2_thread.start()
servo3_thread.start()
servo4_thread.start()

# Wait for all threads to complete
servo1_thread.join()
servo2_thread.join()
servo3_thread.join()
servo4_thread.join()

# Add a sleep to observe the motion
sleep(1)
```

Figure:14,15 (Code for Servo Motors)

```python
from gpiozero import Motor
from time import sleep

motor=Motor(forward=17, backward=27)

try:
    while True:
        motor.forward()
        sleep(2)

        motor.stop()
        sleep(2)

        motor.backward()
        sleep(2)

        motor.stop()
        sleep(2)

except KeyboardInterrupt:
       print("interrupt")

finally:
        motor.stop()
        print("clean")
```

Figure:16 (Code for DC Motor)

```python
import cv2
import requests
import numpy as np

pip install inference-cli && inference server start

inference infer https://source.roboflow.com/cC5SUtHoWhyKgkZhVhf7/0EMUOWG0e3pnv0J2n2aO/original.jp
--api-key ELcwa5yAOcEg67XdglkY \
--project-id smart-bin-detection-system --model-version 3
# Set the URL of your phone's camera stream
stream_url = "http://192.168.102.146:8080/video"  # e.g., "http://192.168.0.100:8080/video"

# Roboflow API details
#ROBOFLOW_API_URL = "https://detect.roboflow.com/smart-bin-detection-system/3"
#API_KEY = "ELcwa5yAOcEg67XdglkY"
IMAGE_SIZE = 416  # Resize images to this resolution for inference

# Open the video stream
cap = cv2.VideoCapture(stream_url)

if not cap.isOpened():
    print("Error: Unable to open video stream")
    exit()

def infer_frame(frame):
    """Send frame to Roboflow for inference and return predictions."""
    # Resize frame to 416x416 for Roboflow API
    resized_frame = cv2.resize(frame, (IMAGE_SIZE, IMAGE_SIZE))

    # Encode image as JPEG
    success, img_encoded = cv2.imencode('.jpg', resized_frame)
    if not success:
        print("Failed to encode frame")
        return None

    # Prepare request data
    img_bytes = img_encoded.tobytes()
    response = requests.post(
        f"{ROBOFLOW_API_URL}?access_token={API_KEY}",
        files={"file": img_bytes},
        headers={"Content-Type": "multipart/form-data"}  # Correct Content-Type for file upload
    )

    # Check if the request was successful
    if response.status_code != 200:
```

```python
        headers={"Content-Type": "multipart/form-data"}  # Correct Content-Type for file up
    )

    # Check if the request was successful
    if response.status_code != 200:
        print(f"Error: Received status code {response.status_code} from Roboflow")
        return None

    # Parse the JSON response from Roboflow
    try:
        predictions = response.json()
    except ValueError:
        print("Error: Failed to parse response as JSON")
        return None

    return predictions

while True:
    ret, frame = cap.read()  # Capture frame-by-frame from the video stream

    if not ret:
        print("Failed to grab frame")
        break

    # Send the frame to Roboflow for object detection
    predictions = infer_frame(frame)

    if predictions and 'predictions' in predictions:
        for pred in predictions['predictions']:
            # Get bounding box and label details
            x, y, w, h = pred['x'], pred['y'], pred['width'], pred['height']
            label = pred['class']  # Waste category (e.g., "organic", "recyclable", etc.)

            # Calculate bounding box coordinates
            start_point = (int(x - w / 2), int(y - h / 2))
            end_point = (int(x + w / 2), int(y + h / 2))

            # Draw the bounding box and label on the frame
            cv2.rectangle(frame, start_point, end_point, (0, 255, 0), 2)
            cv2.putText(frame, label, (start_point[0], start_point[1] - 10),
                        cv2.FONT_HERSHEY_SIMPLEX, 0.9, (36, 255, 12), 2)

    # Show the frame with predictions
    cv2.imshow('Waste Classification', frame)
```

Figure:17,18 (Classification Code)

5.4 SCOPE FOR FUTURE WORK

1. **Handling Mixed Waste:** One of the key challenges identified is the system's inability to efficiently classify waste items that contain mixed materials. Future work could focus on improving the system's ability to deal with such mixed waste, possibly by enhancing the machine learning algorithms.

2. **Scalability:** Although the system is designed for small to medium-scale applications, future work could explore how to scale the system for larger waste management setups, such as city-wide deployments or industrial applications. This would involve ensuring the system's hardware and software can handle higher volumes of waste without compromising performance.

3. **Integration with IoT and Real-Time Monitoring:** The project can be expanded by integrating the system with Internet of Things (IoT) platforms to enable real-time monitoring of waste levels in bins. This could facilitate automated alerts when bins are full and optimize collection schedules, making the entire waste management process more efficient.

4. **Enhanced Data Collection and Machine Learning Models:** Future work could focus on expanding the dataset to include more diverse waste categories and edge cases. This would improve the accuracy of the machine learning model. Moreover, using more advanced models like YOLOv7 or exploring different architectures could enhance the system's performance.
5. **Energy Efficiency and Sustainability:** Further research could focus on reducing the system's energy consumption, making it more suitable for off-grid locations or areas with limited power supply. This could involve optimizing the hardware components or integrating renewable energy sources like solar panels to power the system.
6. **User Behavior and Engagement:** Future work could involve studying how users interact with the system and developing ways to improve compliance and user engagement. For example, adding gamification elements or mobile app integration could encourage better waste disposal habits and increase public awareness.

5.5 PROSPECTIVE LEARNING

1. **Integration of Machine Learning and Computer Vision:**
   - Understanding how to implement machine learning models (like YOLOv5) for real-time object detection and classification.
   - Learning the use of computer vision techniques for image processing and feature extraction with OpenCV.
2. **Hardware-Software Integration:**
   - Gaining experience in integrating hardware components such as Raspberry Pi, servo motors, and camera modules with software systems for real-time control and automation.
   - Understanding the control mechanisms for actuators (servo motors) based on classification results.

3. **IoT and Embedded Systems:**
    - Learning how to implement IoT-based systems using Raspberry Pi and Python, enabling communication between sensors, actuators, and cloud-based data storage or analytics platforms (if integrated).
4. **Data Collection and Preprocessing:**
    - Experience in collecting, annotating, and preprocessing large datasets (using tools like Roboflow) for training machine learning models.
    - Understanding the importance of data augmentation to enhance model robustness.
5. **Waste Management and Environmental Impact:**
    - Learning about sustainable waste management practices and the importance of proper waste segregation in reducing pollution and landfill waste.
6. **Automation in Public Services:**
    - Exploring how automation can be applied to public services like waste management to increase efficiency and reduce human labor.
7. **Collaboration and Teamwork:**
    - Developing skills in team collaboration and project management, particularly in coordinating different aspects like hardware design, software development, and machine learning.
8. **Waste Generation Patterns:**
    - By analyzing data collected from various sensors, the system can learn and predict waste generation patterns. This can help in optimizing waste collection routes and schedules.
9. **Innovation in Waste Segregation:**
    - Significantly improving the recycling processes and reducing the amount of waste that goes into landfills.

5.6 CONCLUSION

In conclusion, the deployment of a smart waste detection system using IoT technology stands as a testament to the innovative strides being made in waste management. It exemplifies how technology can be harnessed to address environmental challenges, enhance urban living, and promote sustainable practices. The learnings from such a system are multifaceted, extending beyond operational efficiencies to encompass community engagement, public health, and policy development.

As we continue to integrate IoT into various aspects of our lives, systems like these will become increasingly crucial in our pursuit of a cleaner, more efficient, and environmentally conscious society. The data-driven insights and real-time monitoring capabilities provided by IoT in waste management are not just incremental improvements but are pivotal in the shift towards a circular economy where waste is minimized, and resources are utilized optimally.

Ultimately, the smart waste detection system is a step forward in realizing the vision of smart cities where technology and data work in harmony to improve the quality of life for all citizens while preserving the planet for future generations. It's a learning curve that paves the way for more informed decisions, responsible behaviors, and innovative solutions to some of the most pressing environmental issues of our time. As illustrated by the Gantt Chart previously, the exact month-wise timeline from the month of January till the month of October has been as follows:

CHAPTER 6
PROJECT TIMELINE

6.1 TIMELINE

1. January: During the month of January, the team focused on defining the objectives of the project. This involves discussing various project ideas, finalising the present idea, establishing clear goals, identifying the scope, and outlining the requirements necessary for the successful completion of the project. This was a group effort with equal contributions from all members i.e, Sheen, Shehjaar, Ananya, Sartaj and Yuvraj.

2. February: In February, the team conduct a thorough literature review. This included researching existing technologies, methods, and systems related to waste segregation, image processing, and machine learning. The purpose was to gather insights and identify gaps that the project can address. This, again, had equal contributions from every member, with each member giving their insights after perusing through 2-3 research papers on their own.

3. March: Data collection took place in March. The team gathered a comprehensive dataset of images representing different types of waste (organic, recyclable, non-recyclable). This dataset has been crucial for training the machine learning model. This was assigned to Sheen, Shehjaar and Sartaj.

4. April: In April, the team began pre-processing of the collected data. This step involved cleaning the data, resizing images, normalizing pixel values, and performing data augmentation to increase the diversity and robustness of the dataset. This also involved rigorous annotation of the collected images on Roboflow. We also kept updating our dataset with new images, which was basically a continuation of our work in March. We did this to increase the accuracy of our model. This work with the dataset was done by Sheen, Shehjaar and Sartaj. In the meantime, two of the team members, Ananya and Yuvraj began work on the development of our software model, i.e, basically the entire backend of our project, which was development and training of our machine learning model.

5. May: Software development for the machine learning model continued in May, while the dataset was continuously being updated by us any time we would come across any sort of new data/image that we believed would improve the training of our ML model. The division of work for these tasks was the same as the month of March and April.

6. June-July: Ananya and Shehjaar focused on refining the Roboflow machine learning model to enhance the accuracy of waste classification. This included improving the dataset, tuning the model, and validating its performance with real waste samples to ensure reliable classification of organic, recyclable, and non-recyclable waste.

7. July-August: Sartaj worked on completing the construction of the wooden box structure. This involved building a robust frame to house the smart bin components, such as the motors, flaps, and compartments for waste segregation. Meanwhile, once the model was fine-tuned, Ananya, Shehjaar, and Yuvraj integrated the Roboflow model with Raspberry Pi, ensuring that the real-time waste classification system would be operational and could process waste inputs as they would be placed in the bin.

8. August-September: Sheen and Sartaj installed the servo motors that control the opening and closing of the bin's flaps. The motors were linked to the waste classification output to direct waste into the correct compartment based on its category. They also configured the DC motor that controls the upper sliding mechanism where the waste is inserted. This mechanism ensures the waste is properly positioned for classification.

9. September: Ananya, Shehjaar, and Yuvraj developed the decision-making logic (if-else mechanism) that controls the flaps. Based on the classification performed by the model, the system directs the waste into the appropriate compartment.

10. September-October: The entire team performed thorough testing of the system, ensuring the accuracy of the waste classification and overall operation. Bugs or performance issues were debugged to improve the system’s reliability and effectiveness. Hence, the project reached its conclusion with the full deployment of the waste segregation bin. The system was tested in real-world conditions, showcasing the smart bin's ability to accurately classify and segregate waste based on machine learning and automated mechanisms

6.2 WORK BREAKDOWN & GANTT CHART

The work breakdown for our project WasteWise till the month of June mainly consisted of the following components:

1. **Project Planning (Defining Objectives):** This included brain-storming sessions with inputs from all the group members, discussing a plethora of ideas and problem statements, and finalising "WasteWise" as the project we would begin work on.
2. **Project Planning (Literature Review):** This included searching for and studying various research papers that have discussed approaches to waste segregation using image processing and machine learning algorithms. Every group member looked up 2-3 research papers for the same, and the result was further definition of our plan for the project thoroughly backed up by sufficient research that allowed us to decrease loopholes and increase the novelty of our work.
3. **Data Collection**: Since the backend of our project works on machine learning algorithms, an essential part of our work has been building an accurate and foolproof dataset that would lead to accurate training of the ML model. For this purpose, we used already-available datasets on the internet and greatly improved on them by adding images of waste items that we photographed ourselves on the university premises, to make our dataset as relevant to our campus as was possible.
4. **Pre-processing of Collected Data:** Data preprocessing was a crucial step in the development of our machine learning model for image classification tasks. This step ensured that our data is clean, formatted correctly, and ready for the model to learn from. It basically included annotating the collected images and carefully classifying them into the three waste categories of waste (as mentioned before): recyclable, non-recyclable and organic.
5. **Model Development (Software):** The software part of our project involved coding the backend in Python using the Roboflow platform, in order to train the model of machine learning to recognise and classify objects based on the pre-processed dataset as mentioned earlier. Convolutional Neural Networks (CNN) were used for the same.

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
1	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓	✓					
4	Pre-processing of Collected Data				✓	✓					
5	Model Development (Software)				✓	✓	✓	✓			
6	Model Development (Hardware)						✓	✓	✓		
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 1: (Overall Gantt Chart)

6.3 INDIVIDUAL GANTT CHART

1. **Individual Gantt Chart for Sheen (102106211):**

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
1	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓						
4	Pre-processing of Collected Data					✓					
5	Model Development (Software)										
6	Model Development (Hardware)						✓	✓	✓		
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 2: (Gantt Chart for Sheen)

2. **Individual Gantt Chart for Shehjaar (102106006):**

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓						
4	Pre-processing of Collected Data				✓	✓					
5	Model Development (Software)				✓	✓	✓	✓			
6	Model Development (Hardware)										
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 3: (Gantt Chart for Shehjaar)

3. Individual Gantt Chart for Ananya (102106137):

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
1	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓						
4	Pre-processing of Collected Data				✓	✓					
5	Model Development (Software)				✓	✓	✓	✓			
6	Model Development (Hardware)										
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 4: (Gantt Chart for Ananya)

4. Individual Gantt Chart for Sartaj (102106223):

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
1	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓						
4	Pre-processing of Collected Data					✓					
5	Model Development (Software)										
6	Model Development (Hardware)						✓	✓	✓		
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 5: (Gantt Chart for Sartaj)

5. Individual Gantt Chart for Yuvraj (102106007):

Sr. No.	Task	January	February	March	April	May	June	July	August	September	October
1	Project Planning (Defining objectives)	✓									
2	Project Planning (Literature Review)	✓	✓								
3	Data Collection (Images)			✓	✓						
4	Pre-processing of Collected Data				✓	✓					
5	Model Development (Software)				✓	✓	✓	✓			
6	Model Development (Hardware)										
7	Integration and Development								✓	✓	
8	Testing and Validation									✓	✓

Table 6: (Gantt Chart for Yuvraj)

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

The development of the "WasteWise: A Smart Segregation Bin" has demonstrated significant potential in transforming waste management practices by automating the waste segregation process. By integrating machine learning algorithms, computer vision, and robotic automation, the system offers a precise, efficient, and cost-effective solution for classifying waste into organic, recyclable, and non-recyclable categories. This automation not only enhances the recycling process but also reduces human intervention and minimizes the risk of contamination in waste streams.

The system's ability to perform real-time waste classification and automatic segregation has proven to be a key factor in improving the efficiency of waste management, contributing to environmental sustainability. The use of affordable hardware, such as Raspberry Pi, and open-source software makes the system accessible and scalable, catering to a wide range of applications, from households to public spaces. Its modular design and ease of deployment ensure that it can be integrated into existing waste management systems with minimal effort.

Overall, the project successfully meets its objectives of creating an intelligent, practical, and sustainable solution for waste segregation. The system's adaptability, low cost, and environmental impact make it a promising tool for large-scale implementation in smart cities and urban environments, contributing to more efficient and sustainable waste management practices.

7.2 FUTURE WORK

1. **Expanded Waste Classification**: Incorporate more waste categories, such as electronic waste, hazardous waste, and medical waste, to further improve sorting precision and efficiency.

2. **Enhanced Machine Learning Models:** Implement more advanced machine learning models, such as YOLOv6 or transformer-based vision models, to improve the speed and accuracy of object detection and classification.

3. **Integration with IoT and Cloud Platforms**: Develop real-time monitoring and data analytics capabilities by integrating the system with IoT platforms and cloud services for remote monitoring and performance tracking.

4. **Energy Efficiency and Power Solutions:** Explore renewable energy sources such as solar power to make the system energy-efficient and self-sustainable, especially for deployment in remote areas.

5. **User Interface and Interaction:** Design a user-friendly mobile or web-based interface that allows users to view waste segregation data, bin status, and receive notifications for bin maintenance or emptying.

6. **Scalability for Smart City Implementation:** Extend the system for large-scale deployment in smart cities, integrating it with municipal waste management systems to automate waste collection and processing across urban areas.

7. **Robotic Arm Integration:** Incorporate robotic arms to handle more complex waste sorting tasks, especially for large and bulky items or mixed waste streams.

These advancements will enhance the system's accuracy, user interaction, and environmental impact, making it an indispensable tool for future smart waste management solutions.

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